



Numerical and experimental study of mixed convection heat transfer and fluid flow characteristics of plate-fin heat sinks



Han-Taw Chen^{a,*}, Hung-Chia Tseng^a, Shih-Wei Jhu^a, Jiang-Ren Chang^b

^a Department of Mechanical Engineering, National Cheng Kung University, Tainan 701, Taiwan

^b Department of Systems Engineering and Naval Architecture, National Taiwan Ocean University, Keelung 202, Taiwan

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ABSTRACT

This study applies three-dimensional computational fluid dynamics (CFD) commercial software along with the inverse method and experimental data to determine the mixed convection heat transfer and fluid flow characteristics of a plate-fin heat sink in a wind tunnel. The inverse method of the finite difference method along with the experimental temperature data is applied to determine the unknown heat transfer coefficient on the fin. Commercial software combined with various flow models is used to obtain air temperature and velocity profiles, heat transfer coefficient on fins, fin surface temperature and pressure drop. More accurate heat transfer and fluid flow characteristics can be obtained by the appropriate flow model and the number of grid points, if the resulting heat transfer coefficient and the fin temperature at each measurement location are close to the inverse results of the heat transfer coefficient and the experimental temperature data, respectively. The interesting finding is that the results obtained by the RNG k - ϵ turbulence model are more accurate than those by the laminar flow model. FLUENT 4 has better accuracy than FLUENT 15 along with standard wall functions and enhanced wall treatment. In addition, the total number of grid points needs to be increased with increasing air velocity and fin spacing. The dimensionless wall distance can vary with air velocity. The pressure drop has a large variation in the specific range of the fin spacing, and the secondary vortices can be found at both corners of the wind tunnel. It is worth mentioning that the strength of the secondary flow decreases with decreasing fin spacing. The effect of the flow model, near-wall treatment, FLUENT version and grid points on the results obtained cannot be ignored. To our knowledge, few researchers have used similar methods to investigate this problem in the open literature. The two proposed correlations are closer to the obtained inverse and numerical results than the existing results.

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1. Introduction

Due to the rapid development of electronic technology, electronic devices are ubiquitous in our lives. When the size and weight of electronic components are reduced, the heat flux per unit area increases dramatically. Under these circumstances, the working temperature of electronic components may exceed the desired temperature level. Thus, promoting the heat transfer rate during the working process at the desired operating temperature plays an important role for insuring a reliable operation of electronic components. This implies that proper heat sink design is attractive for these applications since it provides a more economical solution to the heat issue. Conventional electronic cooling using a heat sink with cooling fans is superior in terms of unit price, weight and

reliability. In order to design a practical heat sink, some criteria, such as a large heat transfer rate, a low pressure drop and a simple structure, should be considered. Heat sink fins increase the heat flow per surface area. However, it is necessary to note the interaction between the local heat transfer and flow distributions within the fins while designing the heat sink. Thus, complex three-dimensional (3D) flow and thermal fields can be observed for various fin geometries. Heat transfer dissipation from an array of parallel rectangular fins on a horizontal surface has been studied for a long time. Although many researchers have applied 3D computational fluid dynamics (CFD) commercial software to obtain the heat transfer and fluid flow characteristics of plate-fin heat sinks (PFHSs) for various air velocities, the accuracy of the results needs to be verified, especially by comparison of reliable results and the experimental temperature measurements. Thus, 3D CFD software along with appropriate flow models and experimental data can help in designing high-performance heat sinks for these devices.

* Corresponding author.

E-mail address: htchen@mail.ncku.edu.tw (H.-T. Chen).

Nomenclature

A_f	lateral surface area of the fin, m^2	Pr_t	turbulent Prandtl number
A_j	area of the j th sub-fin region, m^2	Q	total heat transfer rate dissipated from the fin, W
$[A]$	global conduction matrix	q_j	heat transfer rate dissipated from the j th sub-fin region, W
C	fin tip-to-shroud clearance	Ra	Rayleigh number, $Ra = g\beta(T_0 - T_\infty)H^3/(\nu\alpha)$
c_p	specific heat of air, J/(kg K)	Re	Reynolds number, $V_a t/\nu$
D_h	hydraulic diameter of the fin array, m	Re_d	Reynolds number, $V_a D_h/\nu$
$[K]$	global conduction matrix	Ri	Richardson number, $Ri = RaPr/Re_d^2$
$[F]$	force matrix	S	fin spacing, mm
Gr	Grashof number, $Gr = g\beta(T_0 - T_\infty)D_h^3/\nu^2$	S_{ij}	mean strain rate tensor, $(\partial u_i/\partial x_j + \partial u_j/\partial x_i)/2$
g_j	gravitational acceleration component in the x_j direction, m/s^2	T	fin temperature, K
H	fin height, m	T_a	air temperature, K
h	heat transfer coefficient on the fins, $W/(m^2 K)$	T_j	fin temperature at the j th measurement location, K
$\bar{h}$	average heat transfer coefficient on the fins, $W/(m^2 K)$	T_0	fin base temperature, K
$\bar{h}_b$	heat transfer coefficient based on the fin base temperature, $W/(m^2 K)$	T_∞	ambient air temperature, K
$\bar{h}_j$	average heat transfer coefficient in the j th sub-fin region, $W/(m^2 K)$	$[T]$	global temperature matrix
k	turbulent kinetic energy	t	fin thickness, m
k_a	thermal conductivity of air, $W/(m K)$	u_i	air velocity component in the x_i direction, m/s
k_f	thermal conductivity of fin, $W/(m K)$	V_a	frontal air velocity, m/s
L	fin length, m	x, y, z	Cartesian coordinates, m
ℓ_x	distance between two adjacent nodes in the x direction, $L/(N_x - 1)$	x_i	index symbol of the Cartesian coordinate system, m
ℓ_y	distance between two adjacent nodes in the y direction, $H/(N_y - 1)$	y^*	dimensionless wall distance
N	number of sub-fin regions	Greek symbols	
Nu	Nusselt number, $\bar{h}t/k_a$	α_e, α_k	turbulent Prandtl numbers for diffusion of k and ε
Nu_d	Nusselt number, $\bar{h}_b D_h/k_a$	β	volumetric thermal expansion coefficient
N_t	total number of grid points	β_t	parameter in RNG k - ε turbulence model
N_{tf}	number of grid points on the lateral surface of the fin	δ_{ij}, δ_{j2}	Kronecker delta function
N_x	number of nodes in the x direction	ε	viscous dissipation rate of turbulence kinetic energy
N_y	number of nodes in the y direction	η	mean flow field
p	pressure, Pa	η_f	fin efficiency
p^*	wetted perimeter of the fin array, m	ν	laminar kinematic viscosity, $kg/(s m)$
		ν_{eff}	effective kinematic viscosity, $kg/(s m)$
		ν_t	turbulent kinematic viscosity, $kg/(s m)$
		ρ	air density, kg/m^3

PFHSs have been widely used in cooling electronic devices because they are easy to manufacture, and have simple structure and low cost. Various forms of the PFHSs have been manufactured and supplied to markets in large quantity [1]. Numerical studies and some experiments for the thermal performance of plate-fin and plate-pin-fin heat sinks have been investigated by Yu et al. [1], Kim et al. [2], Li and Chao [3] and Yang and Peng [4]. The thermal resistance of the heat sink is obtained by the ratio of the temperature difference and the heat dissipation power applied to the fin base. This temperature difference is the difference between the maximum temperature at the fin base and the ambient air temperature. El-Sayed et al. [5] varied the fin height, fin width, fin spacing, number of fins and the distance from the fin tip to the shroud to study the performance PFHSs. They concluded that the pressure drop decreased with increasing fin spacing. The mean Nusselt number increases with increasing fin spacing. Under the assumptions of the one-dimensional heat conduction model and constant mean heat transfer coefficient, they proposed the relationship between average Nusselt number and Reynolds number. Dogan and Sivrioglu [6] used the CFD software FLUENT along with the experimental method to investigate the effect of the clearance gap on laminar mixed convection heat transfer from the fin array in the horizontal channel. The numerical results are obtained under the grid independence assumption. Culham and Muzychka [7] applied entropy generation minimization for optimization of PFHSs. Iyengar and Bar-Cohen [8] presented a coefficient of performance analysis for PFHSs in forced convection using least-energy optimization with

the entropy minimization methodology. Najafi et al. [9] studied the energy and cost optimization of plate and fin heat exchangers using genetic algorithm. Xie et al. [10] used the entropy generation minimization theory coupled with constructal law to determine the optimal pin-fin of the heat exchanger.

Velayati and Yaghoubi [11] applied the finite volume method along with the SIMPLE pressure-velocity coupling algorithm to solve 3D turbulent flow and heat transfer characteristics of parallel heated rectangular plates mounted over an insulating base plate. Their results show very complex 3D flow characteristics within parallel bluff plates. This complex flow pattern is accompanied by fluid separation and reattachment. An approximate grid point of $165 \times 88 \times 32$ is used to determine the fluid flow and heat transfer characteristics of parallel heated rectangular thick plates. It is seen from Refs. [4,11] that the solution is substantially independent of the grid for each case. Sparrow et al. [12] used experimental investigation to study heat transfer and pressure drop for airflow in arrays of heat generating rectangular modules deployed along one wall of a flat rectangular duct. Under the assumptions of uniform air velocity and one-dimensional heat flow, Elshafei [13] applied theoretical and experimental studies to investigate the effects of the duct velocity, fin density, tip-to-shroud clearance on flow bypass, thermal performance and pressure drop across a longitudinal aluminum fin array. It is seen from Ref. [14] that a comparison of the numerical predictions and experimental data is made for the hydrodynamic and thermal fields in a two-channel plate heat exchanger using laminar and two-equation turbulence models.

The friction factor, Nusselt number, the outlet temperatures of the two streams and the temperature distributions on the channel end-plates are compared. The results show that the realizable k - ε turbulence model along with the non-equilibrium wall function is used to obtain the closest result compared to the experimental data. Chen et al. [15] applied commercial software [16] combined with laminar and RNG k - ε turbulence models to estimate the heat transfer and fluid flow characteristics of two-row staggered plate fin and tube heat exchangers. Chen et al. [17,18] also used the inverse method and commercial software [16] along with experimental data and various flow models to estimate the natural convection heat transfer characteristics of vertical plate fin and tube heat exchangers with various tube diameters. From Refs. [15,17,18], more accurate results can be obtained, if the obtained heat transfer coefficient and the fin temperature at each measurement location are close to the inverse results and the experimental temperature data, respectively. It is seen from Refs. [17–19] that the numerical results obtained by RNG k - ε turbulence model for the square fins and by the zero-equation turbulence model for the annular fins are closer to the inverse results and match the correlation in Ref. [20] than the other flow models. An interesting finding is that the number of grid points may need to vary with fin spacing and air velocity. The choice of the appropriate flow model can be important to get a closer result compared with the experimental data.

Chen et al. [21] applied the inverse method and commercial software [16] along with the experimental data to determine the heat transfer and fluid flow characteristics for the fins mounted on a horizontal heating plate at air velocity $V_a = 1$ m/s with various fin spacings. It is found that the heat transfer coefficient obtained by the commercial software [16] combined with the RNG k - ε turbulence model is in good agreement with the inverse result and the existing correlation [12]. The resulting fin temperature is also close to the experimental temperature data at each measurement location. The results show that the number of grid points may depend on the fin spacing. The average heat transfer coefficient obtained by the inverse method and commercial software [16] decreases with increasing fin spacing. This phenomenon is caused by the definition of heat transfer coefficient. However, the heat transfer coefficient obtained from the fin base temperature increases with increasing fin spacing. Therefore, this study proposes an alternative definition to determine the average heat transfer coefficient obtained by the inverse method compared with the FLUENT result. It is seen from Ref. [21] that the effects of the air velocity, flow model, FLUENT version, near-wall treatment and grid points on the results obtained are not investigated. Thus, this study uses an inverse method and commercial software [16,22] along with two different flow models and near-wall treatments to determine more accurate heat transfer and fluid flow characteristics of the plate-fin heat sinks installed in the wind tunnel for various air velocities. This study needs to solve the conjugate convection heat transfer in the air flow field and the fin heat conduction equation. The resulting fin temperature and heat transfer coefficient will be compared with the experimental temperature data and the inverse results [21,23]. Two correlations between the Nusselt number and the Reynolds number are proposed based on the obtained inverse results and numerical results.

2. Inverse scheme

The two-dimensional inverse heat conduction problem is introduced to estimate the unknown heat transfer coefficient on the vertical fins attached to the horizontal heating plate for various air velocities and fin spacings. For the convenience of the inverse method, three vertical fins placed on the top of the horizontal heating plate are regarded as plate-fin heat sinks. The fin temperature,

ambient air temperature and air velocity are measured by the experimental apparatus of the wind tunnel, as shown in Fig. 1 of Ref. [21]. The physical geometry of this wind tunnel is shown in Fig. 3 of Ref. [21]. The inverse method of the finite difference method along with the experimental temperature data and the least squares method is used to predict the heat transfer coefficient on the vertical fins attached to the horizontal heating plate. Because of the assumption of non-uniform distribution of unknown heat transfer coefficient, the entire fin is divided into several sub-fin regions. The heat transfer coefficient in each sub-fin region is assumed to be an unknown constant. The physical geometry with the measurement location and the sub-fin region is shown in Fig. 2 of Ref. [21]. To avoid duplication, a detailed diagram of the experimental apparatus and the experimental method can be found in Ref. [21]. Under the assumption of a thin fin, the temperature gradient in the z direction (the fin thickness) can be disregarded. Thus, the fin temperature varies only in the x and y directions. In addition, the surface area of the fin edge is small when compared to the total fin surface area. This implies that the actual heat transfer rate dissipated through the fin tip is rather small compared to the total heat transfer rate drawn from the fin base. Thus, the boundary condition at the edge surface of the fin is assumed to be insulated [15,17–19,21]. Under the assumption of steady state and constant thermal properties, the fin heat conduction equation can be expressed as

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = \frac{2h(x,y)}{k_f t} (T - T_\infty) \quad \text{for } 0 < x < L, \quad 0 < y < H \quad (1)$$

The corresponding boundary conditions are

$$\frac{\partial T}{\partial x} = 0 \quad \text{at } x = 0 \quad \text{and } L \quad (2)$$

$$T(x, 0) = T_0 \quad (3)$$

and

$$\frac{\partial T}{\partial y} = 0 \quad \text{at } y = H \quad (4)$$

where x and y are the Cartesian coordinates. L , H and t denote the length, height and thickness of the fin, respectively. The parameter $h(x, y)$ is an unknown heat transfer coefficient. k_f is the thermal conductivity of the fins. T_0 and T_∞ denote the fin base temperature and the ambient air temperature, respectively.

The finite difference form of Eq. (1) in the k th sub-fin region can be expressed as

$$\frac{T_{i+1,j} - 2T_{i,j} + T_{i-1,j}}{\ell_x^2} + \frac{T_{i,j+1} - 2T_{i,j} + T_{i,j-1}}{\ell_y^2} = \frac{2\bar{h}_k}{k_f t} T_{i,j} \quad \text{for} \quad (5)$$

$$i = 1, 2, \dots, N_x, \quad j = 1, 2, \dots, N_y \quad \text{and} \quad k = 1, 2, \dots, N$$

where N_x and N_y are, respectively, the number of nodes in the x and y directions. ℓ_x and ℓ_y are defined as $\ell_x = L/(N_x - 1)$ and $\ell_y = H/(N_y - 1)$. N is the number of sub-fin regions.

The difference equations of the nodes at the boundary, the interface between two adjacent sub-fin regions and the intersection of four adjacent sub-fin regions are similar to those shown in Ref. [21]. To avoid duplication, they are not reiterated in this manuscript.

The rearrangement of Eq. (5) with its corresponding difference equation generates the following matrix equation as

$$[K][T] = [F] \quad (6)$$

where $[K]$ is the global conduction matrix. $[T]$ is a matrix representing the node temperature. $[F]$ is the force matrix. The fin temperature at the selected measurement locations is obtained from Eq. (6) using Gaussian elimination.

In order to estimate the unknown heat transfer coefficient $\bar{h}_j$ in the j th sub-fin region, additional information of the fin temperature measured at N locations may be required. The more the number of sub-fin regions, the more accurate the estimated heat transfer coefficient is. An obtrusive thermocouple on the fin may significantly interfere with the flow and heat transfer behaviors between the two adjacent fins and the heat transfer behavior on the fins. Thus, the T -type thermocouple is applied to measure the fin temperature only at selected measurement locations and ambient air temperature. The accuracy of the T -type thermocouple is $\pm 0.4\%$.

The least squares minimization technique can be applied to minimize the sum of squares of the deviations between the calculated and measured fin temperatures at the selected locations. The error in estimating $E(\bar{h}_1, \bar{h}_2, \dots, \bar{h}_N)$ is minimized and is defined as

$$E(\bar{h}_1, \bar{h}_2, \dots, \bar{h}_N) = \sum_{j=1}^N [T_j^{inv} - T_j^{mea}]^2 \quad (7)$$

where T_j^{mea} and T_j^{inv} denote the experimental measured and calculated fin temperatures at the j th measurement location, respectively. T_j^{inv} is determined from Eq. (6).

The estimated values of $\bar{h}_j$ for $j = 1, 2, \dots, N$ can be determined provided that the value of $E(\bar{h}_1, \bar{h}_2, \dots, \bar{h}_N)$ is minimized. The details of estimating the unknown value $\bar{h}_j$ can be found in Refs. [17–19,21]. The computational procedures of this study are repeated until the values of $\left| \frac{T_j^{mea} - T_j^{inv}}{T_j^{mea}} \right|$ for $j = 1, 2, \dots, N$ are all less than 10^{-5} . Once the value of $\bar{h}_j$ is determined for $j = 1, 2, \dots, N$, the average heat transfer coefficient $\bar{h}$, heat transfer coefficient based on the fin base temperature $\bar{h}_b$ and fin efficiency η_f are determined.

3. 3D numerical analysis

Understanding the details of the local heat transfer and fluid flow distribution between the fins is important for heat sink design. Many researchers [1–8,11–13,21,23–27] used various numerical methods to investigate the heat transfer characteristics of plate-fin heat sinks. However, few researchers have investigated the accuracy of numerical results obtained from commercial software along with the inverse results of heat transfer coefficient and fin temperature data at selected measurement locations in the open literature. Therefore, this study further applies commercial software [16,22] along with laminar and RNG k - ε turbulence models to determine the heat transfer and fluid flow characteristics of plate-fin heat sinks mounted on a horizontal heating plate for various air velocities and fin spacings. This study needs to solve the differential equations in the air flow field and the fin heat conduction equation simultaneously. The air velocity pattern, the air temperature profile, the heat transfer coefficient on the fins, the pressure drop and the fin surface temperature distribution are obtained.

A half section of a fin is selected for FLUENT computation. Under the assumption of steady-state and constant thermal properties, the 3D fin heat conduction equation is expressed as

$$\frac{\partial^2 T}{\partial x_i^2} = 0 \quad \text{for} \quad 0 < x < L, \quad 0 < y < H, \quad 0 < z < t/2 \quad (8)$$

In addition to Eqs. (2)–(4), the boundary conditions at $z = 0$ and $t/2$ can be expressed as

$$-k \frac{\partial T}{\partial z} = h(T - T_\infty) \quad \text{at} \quad z = t/2 \quad (9)$$

and

$$\frac{\partial T}{\partial z} = 0 \quad \text{at} \quad z = 0 \quad (10)$$

where the index symbol of the Cartesian coordinate system x_i , $i = 1, 2, 3$, respectively indicate x , y and z . The parameter h represents the local heat transfer coefficient on the fin.

3.1. Different flow models

The air flow with constant properties is assumed to be 3D, incompressible, steady and non-viscous dissipation. Due to higher fin base temperature compared to ambient air temperature, the buoyancy effect is considered. In order to study the effects of the flow model and near-wall treatment on the results obtained, the laminar flow and RNG k - ε turbulence models are introduced. The continuity, momentum and energy equations can be expressed in tensor form for both flow models.

3.1.1. Laminar model

$$\frac{\partial u_i}{\partial x_i} = 0 \quad (11)$$

$$u_j \frac{\partial u_i}{\partial x_j} = -\frac{1}{\rho} \frac{\partial p}{\partial x_i} + \nu \frac{\partial^2 u_i}{\partial x_j^2} + g_j \beta \delta_{j2} (T_a - T_\infty) \quad (12)$$

and

$$c_p u_j \frac{\partial T_a}{\partial x_j} = \frac{k_a}{\rho} \frac{\partial^2 T_a}{\partial x_j^2} \quad (13)$$

3.1.2. RNG k - ε turbulence model

$$u_j \frac{\partial u_i}{\partial x_j} = -\frac{1}{\rho} \frac{\partial p}{\partial x_i} + \nu \frac{\partial}{\partial x_j} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) - \frac{\partial \overline{u'_i u'_j}}{\partial x_j} + g_j \beta \delta_{j2} (T_a - T_\infty) \quad (14)$$

and

$$c_p u_j \frac{\partial T_a}{\partial x_j} = \frac{k_{eff}}{\rho} \frac{\partial^2 T_a}{\partial x_j^2} - c_p \frac{\partial \overline{u'_j T'_a}}{\partial x_j} + 2\nu_{eff} \frac{\partial}{\partial x_j} (u_i S_{ij}) \quad (15)$$

where the continuity equation for RNG k - ε turbulence model is the same as Eq. (11). The parameters u_i , p , g_j and T_a are the velocity component, the pressure, the gravitational acceleration component in the x_j direction and the air temperature, respectively. β is the volumetric thermal expansion coefficient. δ_{j2} is the Kronecker delta. The parameters ρ , ν , c_p and k_a are, respectively, density, laminar kinematic viscosity, specific heat and thermal conductivity of the air. The effective thermal conductivity k_{eff} and the effective kinematic viscosity ν_{eff} are defined as $k_{eff} = k + k_t$ and $\nu_{eff} = \nu + \nu_t$, respectively. The turbulent eddy viscosity ν_t is defined as $\nu_t = c_\mu k^2 / \varepsilon$. The Reynolds stress tensor and the turbulent heat flux combined with Boussinesq approximation are expressed as

$$-\overline{u'_i u'_j} = 2\nu_t S_{ij} - \frac{2}{3} k \delta_{ij} \quad (16)$$

and

$$-\overline{u'_j T'_a} = \frac{c_p \nu_t}{Pr_t} \frac{\partial T_a}{\partial x_j} \quad (17)$$

where the mean strain rate tensor S_{ij} is defined as $S_{ij} = (\partial u_i / \partial x_j + \partial u_j / \partial x_i) / 2$. Pr_t is the turbulent Prandtl number.

The k and ε equations for the RNG k - ε turbulence model with buoyancy effect are expressed as:

$$u_j \frac{\partial k}{\partial x_j} = \frac{\partial}{\partial x_j} \left(\alpha_k v_{eff} \frac{\partial k}{\partial x_j} \right) + \frac{G_k + G_b}{\rho} - \varepsilon \quad (18)$$

and

$$u_j \frac{\partial \varepsilon}{\partial x_j} = \frac{\partial}{\partial x_j} \left(\alpha_\varepsilon v_{eff} \frac{\partial \varepsilon}{\partial x_j} \right) + c_{1\varepsilon} \frac{\varepsilon}{\rho k} (G_k + C_{3\varepsilon} G_b) - \frac{c_{2\varepsilon} \eta^3 (1 - \eta/\eta_0)}{1 + \beta_t \eta^3} \frac{\varepsilon^2}{k} - c_{2\varepsilon} \frac{\varepsilon^2}{k} \quad (19)$$

where the parameters G_k , G_b and η are the production of turbulent kinetic energy due to the velocity gradient, the production of turbulent kinetic energy due to the buoyancy and the ratio of characteristic time scale of turbulence and the mean flow field, respectively. The definitions of G_k , G_b and η can refer to Refs. [3,4,15–18,21,23]. The parameters α_k , α_ε , Pr_t , $c_{1\mu}$, $c_{1\varepsilon}$, $c_{2\varepsilon}$, η_0 and β_t are given as $\alpha_k = \alpha_\varepsilon = 1.393$, $Pr_t = 0.8$, $c_{1\mu} = 0.0845$, $c_{1\varepsilon} = 1.42$, $c_{2\varepsilon} = 1.68$, $\eta_0 = 4.38$ and $\beta_t = 0.012$. The terms $c_{1\varepsilon} G_k \varepsilon / k$ and $c_{2\varepsilon} \rho \varepsilon^2 / k$ represent, respectively, the shear generation and viscous dissipation of ε . The parameter $c_{3\varepsilon}$ is 0 due to the air velocity perpendicular to the direction of gravity.

3.2. Computational procedure

The computational domain is meshed using only the hexahedral mesh generation algorithm. This commercial software uses the finite volume discretization technique to solve the 3D governing differential equations for laminar and turbulent flows. The volume cells for velocity components are staggered with respect to the main control volume cells using the SIMPLEC pressure-velocity coupling algorithm [11].

3.3. Boundary conditions

The matching conditions of the temperature and heat flux at the interface between fins and air can be written as

$$T = T_a \quad \text{and} \quad k_f \frac{\partial T}{\partial z} = k_a \frac{\partial T_a}{\partial z} \quad (20)$$

The average heat transfer coefficient on the fin $\bar{h}$ can be defined as [15,17–19,21]

$$\bar{h} = \frac{\int_{A_f} (T - T_\infty) h(x, y) dA}{\int_{A_f} (T - T_\infty) dA} \quad (21)$$

and

$$\bar{h} = \frac{\int_{A_f} \bar{h} dA}{A_f} \quad (22)$$

where the lateral surface area of the plate fin A_f is defined as $A_f = LH$.

The average heat transfer coefficient $\bar{h}$ is approximated as described in Refs. [15,17,18]

$$\bar{h} \approx \sum_{j=1}^N \bar{h}_j A_j / A_f \quad \text{for inverse method} \quad (23)$$

and

$$\bar{h} \approx \sum_{j=1}^{N_f} h_j / N_f \quad \text{for commercial software} \quad (24)$$

where h_j and N_f represent the heat transfer coefficient at the j th grid point and the number of grid points in the lateral surface of the fin, respectively. A_j is the area of j th sub-fin region.

Due to the limited use of thermocouples, Eqs. (21) and (22) may not be suitable for this study. It may be slightly complicated in

order to obtain the values of $\bar{h}$ and Q_j from Eq. (23). Thus, the $\bar{h}$ and Q_j values obtained from the inverse method can be approximated as

$$\bar{h} \approx \sum_{j=1}^N \bar{h}_j (T_j - T_\infty) A_j / \sum_{j=1}^N (T_j - T_\infty) A_j \quad (25)$$

and

$$Q_j \approx 2A_j \bar{h}_j (T_j^{mea} - T_\infty) \quad \text{for } j = 1, 2, \dots, N \quad (26)$$

The total heat transfer rate dissipated from a fin to the ambient air Q can be expressed as

$$Q = \sum_{j=1}^N Q_j = 2 \sum_{j=1}^N \bar{h}_j \int_{A_j} (T - T_\infty) dA \quad \text{for inverse method} \quad (27)$$

and

$$Q = 2 \int_{A_f} h(x, y) (T - T_\infty) dA \quad \text{for commercial software} \quad (28)$$

Another form of Q can be expressed as

$$Q = 2A_f \bar{h}_b (T_0 - T_\infty) = 2A_f \bar{h} (T_{ave} - T_\infty) \quad (29)$$

where T_{ave} denotes the average of all the fin temperatures at grid points in the lateral surface of the fin. The values of $\bar{h}$ and $\bar{h}_b$ are obtained by Eq. (29).

The fin efficiency η_f can be defined as the ratio of the actual total heat transfer rate from the fin to the dissipated heat from the fin maintained at the fin base temperature T_0 and can be expressed as

$$\eta_f = \frac{Q}{2A_f (T_0 - T_\infty) \bar{h}} = \frac{\bar{h}_b}{\bar{h}} \quad (30)$$

The parameters k_{in} and ε_{in} are defined as inlet turbulence kinetic energy and viscous dissipation rate, respectively. They are the intrinsic values of the commercial software FLUENT and are set to 0.5. Only the half-section of the fins is selected for all calculations. The computational domain is a half of the wind tunnel, as shown in Fig. 1 in dashed lines. The gradient of all dependent variables at the boundary of the wind tunnel is set to zero. The inlet boundary of the computational domain is located $1.5L$ away from the leading edge of the fins. At the inlet boundary, the ambient air is assumed to have uniform velocity V_a and constant temperature T_∞ . Other velocity components in the y and z directions are set to zero. At the solid surface, no-slip conditions are specified. The air temperature at the boundary of the wind tunnel is assumed to be T_∞ . The fin temperature data at selected measurement locations, T_∞ and T_0 are measured from this experiment. The fin base temperature T_0 is constant. In order to obtain the desired numerical results, the downstream outlet boundary away from the rear edge of the test fins may vary with the fin spacing and air speed. The outlet boundary of the computational domain is located $5.0L$ away from the rear edge of the fins for $V_a = 3$ m/s. The stream-wise gradients are set to zero for all dependent variables. The pressure at the outlet boundary is set to the static pressure.

4. Experimental apparatus

This experiment is conducted in a small wind tunnel. The schematic diagram of the experimental apparatus is shown in Fig. 3 of Ref. [21]. The length L_e , height H_e and width W_e of the wind tunnel are 2.26 m, 0.22 m and 0.22 m, respectively. Airflow is straightened by a flow straightener installed in the air inlet of the wind tunnel. The details of the experimental procedures and methods can be

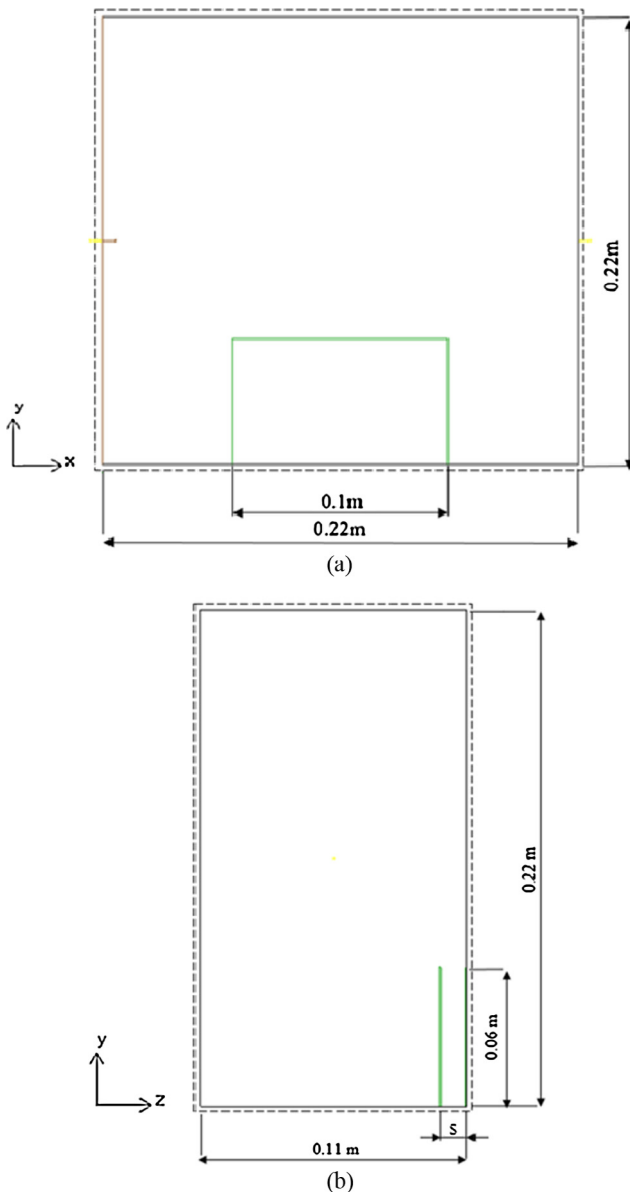


Fig. 1. Schematic diagram of the physical model and the computational domain in dashed lines. (a) x - y plane, (b) y - z plane.

found in Ref. [21]. An anemometer installed at 0.2 m in front of the airflow entering the test fins is used to measure frontal air velocity. The horizontal plate with 0.1 m in length, 0.1 m in width and 0.006 m in thickness and the test fins with 0.1 m long, 0.1 m wide and 0.001 m thick are made from AISI 304 stainless material. The ambient air temperature and the fin temperature are measured using the T -type thermocouple. The accuracy of the T -type thermocouple is $\pm 0.4\%$ for temperature from 0°C to 350°C . A schematic diagram of three parallel vertical rectangular fins mounted on a horizontal heating plate is shown in Fig. 1 of Ref. [21]. In order to heat the three parallel fins, a square 40 W heater with 0.08 m in length is affixed to the bottom of the plate. The test fins and horizontal plate are surrounded with insulating material and heated for about 7600 s. The four thermocouples placed in the gap between the fins and the horizontal plate are fixed to $(L/5, 0)$, $(2L/5, 0)$, $(3L/5, 0)$ and $(4L/5, 0)$. The average of the four measured temperatures is regarded as the fin base temperature T_0 . The thermal contact resistance between the fin and the horizontal plate is assumed to be negligible. The ambient air temperature T_∞ is

measured at 0.3 m in front of the air flow into the test fin as shown in Fig. 3 of Ref. [21].

5. Typical grid distribution

An unstructured grid system containing a non-uniform distribution of the grid points is used to determine all of the numerical results. The mesh density is controlled to ensure appropriate results with various values of air velocity and fin spacing. A finer mesh is needed in some regions where the temperature and velocity gradients are high, such as in the vicinity of the tube and fins. In some regions where the temperature and velocity gradients are small, a coarser mesh can be used to save computing time. In other words, the grid size is non-equal interval. The grid system used in this study is shown in Fig. 2. The number of grid points in the middle fins in the x , y and z directions, respectively, is taken as 4, 9 and 3 with $V_a = 1$ m/s and 12, 9 and 3 with $V_a = 2$ m/s. The number of grid points in the middle fins is taken as 13 and 9 in the x and y directions, respectively, and from 3 to 4 in the z direction for $V_a = 3$ m/s. The number of grid points between two adjacent fins in the x and y directions is the same as the number of intermediate fins for $1 \text{ m/s} \leq V_a \leq 3 \text{ m/s}$, respectively. The number of grid points between two adjacent fins in the z direction is taken from 3 to 131 with $V_a = 1$ m/s, from 5 to 93 with $V_a = 2$ m/s, and from 6 to 94 with $V_a = 3$ m/s. The number of grid points outside the fins in the x and y directions is taken from 25 to 30 and from 55 to 220, respectively, and 22 in the z direction. The various grid points are tested for each case until the obtained fin temperature and heat transfer coefficient estimates are as close as possible to the experimental temperature data, the inverse results and the existing correlations.

6. Results and discussion

All physical properties are evaluated at the film temperature or the average of the fin base temperature and the frontal air temperature. All calculations are performed with the fin height $H = 0.06$ m, the tip-to-shroud clearance $C = 0.16$ m, $t = 0.001$ m, $L = 0.1$ m and $k_f = 14.9 \text{ W/(m K)}$. The width W_D and height H_D of the wind tunnel are $W_D = H_D = 0.22$ m. The width W_{bp} and thickness t_{bp} of the base plate are $W_{bp} = 0.1$ m and $t_{bp} = 0.006$ m. The parameters $N_x = 21$ and $N_y = 17$ are used for the inverse method. The initial guess of the unknown heat transfer coefficient is taken as unity for the inverse method. The rectangular fin is divided into eight regions, i.e., $N = 8$. Eight thermocouples on the fin are fixed at the same position as shown in Fig. 2 of Ref. [21]. Two different versions of FLUENT [16,22] along with the near-wall treatment, experimental temperature measurements and inverse results are used to estimate the

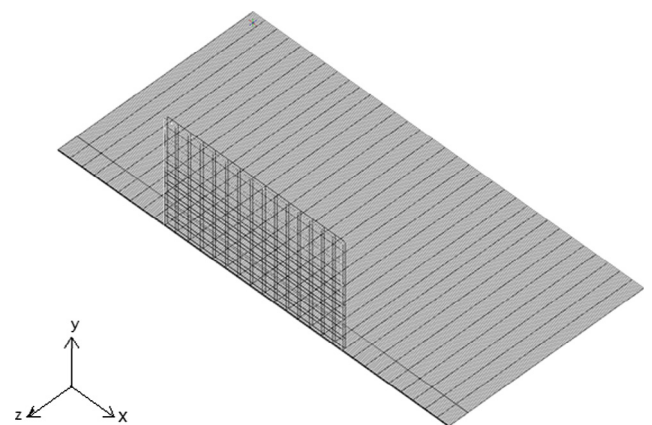


Fig. 2. Typical grid distribution in the computational domain.

fluid flow and heat transfer characteristics of this problem for various air velocities and fin spacings. A suitable flow model and grid points are selected to obtain more accurate results.

Various heat transfer correlations have been proposed for this problem [6,8,11–13,23]. Among these heat transfer correlations, the correlations given by Sparrow et al. [12], Elshafei [13] and Zhou et al. [24] are selected to compare with the results obtained.

Sparrow et al. [10] used experimental investigation to study heat transfer and pressure drop for air flow into the array of heat generating rectangular modules deployed along one wall of a flat rectangular duct. Based on correlations between the Nusselt number Nu and the Reynolds number Re proposed by Velayati and Yaghoubi [11] and Sparrow et al. [12], Chen et al. [21] proposed a similar correlation along with Eqs. (23) and (24) at $V_a = 1.0$ m/s as

$$Nu = 0.0667Re^{0.72} \quad (31)$$

where Nu and Re are defined as $Nu = \bar{h}t/k_a$ and $Re = V_a t/\nu$. It is seen that the effect of the fin spacing is not considered in Eq. (31).

Under the assumption of one-dimensional heat flow, Elshafei [13] proposed a correlation between Nusselt number Nu_d and Reynolds number Re_d based on the hydraulic diameter for $L = 0.225$ m, $H = 0.051$ m, $t = 0.002$ m, $S = 15$ mm, $0 \leq C/H \leq 0.89$, $0 \leq C/S \leq 3$ and $3000 \leq Re_d \leq 38,000$ as

$$Nu_d = \frac{7.522Re_d^{0.182}}{[(1 + C/H)(1 + C/S)]^{0.1096}} \quad (32)$$

where the parameter S represents the fin spacing. Nu_d and Re_d are defined as $Nu_d = \bar{h}_b D_h / k_a$ and $Re_d = V_a D_h / \nu$. The hydraulic diameter of the fin array D_h is defined as $D_h = 4A_F / p^*$. Based on the definition proposed by Elshafei [13], the free-flow area A_F and the wetted perimeter p^* of the fin array are $A_F = 0.04762$ m² and $p^* = 1.24$ m. Thus, the D_h value is 0.1536.

To modify the correlation proposed by Sparrow et al. [12], Elshafei [13] and Chen et al. [21], two correlations are presented as

$$Nu = 0.14Re^{0.57}(0.31 + 47.0S) \quad (33)$$

and

$$Nu_d = \frac{7.522Re_d^{0.19}}{[(1 + C/H)(1 + C/S)]^{0.1096}} [0.6 + (S/0.005)^{0.5}] \quad (34)$$

Zhou et al. [24] proposed a correlation based on the results obtained by the SIMPLEC solution algorithm along with the k - ε turbulence model and optimization procedure for aluminum plate-fin heat sinks, $L = 0.022$ m, $H = 0.012$ m and $L_e = 0.251$ m as

$$Nu = 11.2 \left(\frac{S}{L_e} \right)^{0.6} \left(\frac{t}{L_e} \right)^{0.059} Re_d^{0.64} \quad (35)$$

The value of η_f is obtained by Eqs. (30), (33) and (34).

The limits between the forced convection regime, the mixed convection regime and the free convection regime depend on the geometry of the flow and can be characterized by critical value of the Richardson number Ri , where Ri is defined as $Ri = RaPr/Re_d^2$. The Rayleigh number Ra and the Prandtl number Pr are defined as $Ra = g\beta(T_0 - T_\infty)H^3/\nu\alpha$ and $Pr = \nu/\alpha$. The Richardson number Ri describes the relative importance of free convection to forced convection. The effect of forced convection is dominant for small Ri values. However, free convection is more important for large Ri values. Maughan and Incropera [25,26] used experimentally and numerically to investigate the fully developed laminar mixed convection heat transfer with longitudinal fins in a horizontal parallel plate channel. It is seen from Ref. [25] that a converged one-vortex solution is observed at $Ra = 10,000$. The two-vortex result is obtained for $Ra < 4000$. The buoyancy causes a secondary flow in the form of longitudinal vortices which can

further increase heat transfer. The strength of the secondary flow decreases as the fin spacing decreases.

The comparison of the results obtained by FLUENT 15 [22] along with the RNG k - ε turbulence model is shown in Table 1 with $S = 5$ mm for two different near-wall treatments and V_a values. Near-wall treatment in FLUENT 4 is built-in. The dimensionless wall distance y^+ obtained by FLUENT 15 along with the standard wall functions (SWF) and enhanced wall treatment (EWT) is respectively 8.37 and 16.8 for $V_a = 1$ m/s and 5.81 and 7.00 for $V_a = 3$ m/s. An interesting finding from Table 1 is that the value of y^+ can vary with the air velocity. The value of y^+ obtained by EWT is greater than that by SWF. As can be seen from Tables 1, the difference in $\bar{h}$ and $\bar{h}_b$ values obtained by FLUENT 15 along with SWF and EWT is more pronounced. However, the difference in fin surface temperature and Q value is less obvious. This means that the comparison of the heat transfer coefficient can be important for selecting the appropriate flow model and near-wall treatment.

The results shown in Tables 2–5 and Figs. 3–11 are obtained by FLUENT 4 [16]. It is seen in Ref. [21] that the value of $\bar{h}$ is obtained from Eq. (23) for the inverse method and Eq. (24) for the commercial software [16]. It is found from Eqs. (21)–(24) that the definition of $\bar{h}$ between them is slightly different. Table 2 shows the comparison of $\bar{h}$ and $\bar{h}_b$ between the present results obtained from the commercial software [16] along with the RNG k - ε turbulence model and those given by Sparrow et al. [12], Elshafei [13], Chen et al. [21] and Zhou et al. [24] at $V_a = 1$ m/s and for various values of S . It is seen from Ref. [21] that the $\bar{h}$ value obtained by the commercial software [16] and the inverse method decreases with increasing S value. However, the value of $\bar{h}$ obtained by commercial software [16] and inverse method increases with increasing S value in the present study. The main difference between them can be caused by the definition of $\bar{h}$. This means that the $\bar{h}$ value obtained by Eqs. (23) and (24) cannot be appropriate. Therefore, the definition of $\bar{h}$ needs to be carefully selected. Table 2 show that the difference between $\bar{h}$ values obtained by Eqs. (25) and (29) is small. However, the process of calculating the value of $\bar{h}$ from Eq. (25) is slightly more complex than that from Eq. (29). Thus, the inverse result of the $\bar{h}$ value is obtained from Eq. (29) for comparison with the result obtained from commercial software [16]. The obtained $\bar{h}_b$ and Q results are slightly less than those given by Chen et al. [21], especially for $S = 15$ mm and 20 mm, due to the differences in the number of grid points and the computational domain. The $\bar{h}_b$ and Q values obtained by commercial software [16] are in good agreement with those by the inverse method. The $\bar{h}_b$ and Q values obtained by the commercial software [16] and the inverse method increase with increasing S value.

Table 1

Comparison of the results obtained with $S = 5$ mm for various near-wall treatments and V_a values.

	$V_a = 1$ m/s		$V_a = 3$ m/s	
	EWT	SWF	EWT	SWF
T_1 (K)	310.52	310.29	304.52	304.71
T_2 (K)	312.92	313.25	305.69	306.54
T_3 (K)	304.17	303.67	300.75	300.94
T_4 (K)	306.57	306.49	301.73	302.49
T_5 (K)	301.02	300.42	299.36	299.44
T_6 (K)	302.95	302.48	299.97	300.43
T_7 (K)	299.53	298.93	298.86	298.88
T_8 (K)	301.08	300.31	299.24	299.49
$\bar{h}$ W/m ² K)	16.80	19.50	39.01	35.8
$\bar{h}_b$ W/(m ² K)	4.63	4.41	6.46	6.06
Q (W)	1.48	1.41	1.19	1.12
y^+	16.8	8.37	7.00	5.81

Table 2Comparison of $\bar{h}$ and $\bar{h}_b$ at $V_a = 1$ m/s for various values of S .

S (mm)		5	10	15	20
$\bar{h}_j$ W/(m ² K)	Inv	$\bar{h}_1 = 39.90$	$\bar{h}_1 = 45.49$	$\bar{h}_1 = 86.20$	$\bar{h}_1 = 85.13$
		$\bar{h}_2 = 12.72$	$\bar{h}_2 = 21.36$	$\bar{h}_2 = 43.00$	$\bar{h}_2 = 45.43$
		$\bar{h}_3 = 24.40$	$\bar{h}_3 = 20.04$	$\bar{h}_3 = 38.01$	$\bar{h}_3 = 42.84$
		$\bar{h}_4 = 11.10$	$\bar{h}_4 = 5.76$	$\bar{h}_4 = 19.28$	$\bar{h}_4 = 17.35$
		$\bar{h}_5 = 91.43$	$\bar{h}_5 = 62.30$	$\bar{h}_5 = 18.20$	$\bar{h}_5 = 49.40$
		$\bar{h}_6 = 19.77$	$\bar{h}_6 = 47.18$	$\bar{h}_6 = 13.23$	$\bar{h}_6 = 28.00$
		$\bar{h}_7 = 105.99$	$\bar{h}_7 = 67.56$	$\bar{h}_7 = 59.09$	$\bar{h}_7 = 10.55$
		$\bar{h}_8 = 23.47$	$\bar{h}_8 = 35.57$	$\bar{h}_8 = 15.02$	$\bar{h}_8 = 4.03$
$\bar{h}$ W/(m ² K)	Inv.	Present	24.13	28.98	46.05
		Ref. [21]	41.10	38.16	36.50
	FLUENT 4	Present	22.19	27.49	43.18
		Ref. [21]	38.91	37.22	34.74
		Ref. [12]	34.02	34.04	33.93
$\bar{h}_b$ W/(m ² K)	Inv.	Present	6.71	7.53	10.30
		Ref. [21]	6.77	8.12	11.60
	FLUENT 4	Present	6.85	8.26	11.54
		Ref. [21]	6.66	7.17	13.52
		Ref. [13]	6.68	7.18	7.51
Q (W)	Inv.	2.16	2.15	3.15	3.34
	FLUENT 4	2.07	2.36	2.98	3.21

Table 3Comparison of $\bar{h}$ and $\bar{h}_b$ between inverse and FLUENT 4 results with $d_0 = 27$ mm and various values of V_a and S .

V_a (m/s)	S (mm)	Re	$\bar{h}$ W/(m ² K)		$\bar{h}_b$ W/(m ² K)		Eq. (34)	N_t
			Present results		Present results			
			Inverse	RNG	Inverse	RNG		
1	5	59.15	24.13	22.19	6.71	6.85	6.64	61,479
	10	59.29	28.98	27.49	7.53	8.26	9.35	712,80
	15	58.54	46.05	43.18	10.30	11.54	11.55	546,728
	20	58.27	47.80	47.12	10.41	11.89	11.70	693,924
2	5	118.91	33.95	30.71	8.30	8.37	7.53	278,124
	10	120.12	41.80	41.90	8.86	10.70	10.60	372,240
	15	119.74	55.18	55.47	12.23	12.68	13.11	693,924
	20	118.69	76.96	64.33	13.08	13.10	15.30	1,485,990
3	5	181.72	49.92	43.24	9.90	10.49	8.15	319,968
	10	181.56	52.61	54.73	10.11	11.90	11.47	594,792
	15	180.75	82.12	83.37	13.25	14.80	14.18	857,925
	20	179.86	95.22	89.26	14.26	15.92	16.55	1,519,048

Table 4Comparison of fin temperature and $\bar{h}$ with two different flow models, $S = 5$ mm and various V_a values.

	$V_a = 2$ m/s, $Ra = 4.19 \times 10^5$ $T_0 = 319.71$ K, $T_\infty = 298.77$ K FLUENT 4		Exp. data [23]	$V_a = 3$ m/s, $Ra = 3.10 \times 10^5$ $T_0 = 313.86$ K, $T_\infty = 298.43$ K FLUENT 4		Exp. data [23]
	Laminar	RNG		Laminar	RNG	
T_1 (K)	309.55	308.01	309.64	305.72	304.46	305.53
T_2 (K)	311.48	310.25	313.09	307.24	306.16	307.78
T_3 (K)	304.31	302.88	303.13	301.82	300.77	300.17
T_4 (K)	306.45	304.85	305.47	303.40	302.16	302.13
T_5 (K)	301.72	300.64	300.05	300.05	299.35	298.80
T_6 (K)	303.56	301.96	301.32	301.29	300.20	300.03
T_7 (K)	300.53	299.71	298.99	299.30	298.83	298.48
T_8 (K)	302.03	300.51	299.44	300.23	299.33	298.99
$\bar{h}$ W/(m ² K)	22.59	32.97	33.95 (Inv.)	29.01	43.24	49.92 (Inv.)

The comparison of $\bar{h}$ and $\bar{h}_b$ obtained by commercial software [16] along with the RNG k - ε turbulence model, the inverse method and the proposed correlation (35) is shown in Table 3 for various values of S and V_a . Table 3 shows that the $\bar{h}$ and $\bar{h}_b$ results obtained by the commercial software [16] along with the RNG k - ε

turbulence model are in good agreement with the inverse values for $1 \text{ m/s} \leq V_a \leq 3 \text{ m/s}$ and $5 \text{ mm} \leq S \leq 20 \text{ mm}$. The obtained $\bar{h}$ and $\bar{h}_b$ results increase with increasing air velocity and fin spacing. It is found in Table 2 that the $\bar{h}_b$ and $\bar{h}$ results obtained by the commercial software [16] and the inverse method deviate from those

Table 5
Comparison of fin temperature and $\bar{h}$ with two different flow models, $S = 10$ mm and various V_a values.

	$V_a = 2$ m/s, $Ra = 3.62 \times 10^5$ $T_0 = 316.58$ K, $T_\infty = 298.51$ K FLUENT 4		Exp. data [23]	$V_a = 3$ m/s, $Ra = 3.06 \times 10^5$ $T_0 = 313.95$ K, $T_\infty = 298.65$ K FLUENT 4		Exp. data [23]
	Laminar	RNG		Laminar	RNG	
T_1 (K)	307.77	305.70	307.62	305.87	304.09	305.62
T_2 (K)	309.43	307.33	309.96	307.37	304.97	307.68
T_3 (K)	303.22	301.38	300.85	302.01	300.53	300.22
T_4 (K)	305.08	302.79	302.93	303.56	301.19	301.97
T_5 (K)	301.01	299.67	299.29	300.24	299.30	298.97
T_6 (K)	302.58	300.56	300.21	301.46	299.66	300.16
T_7 (K)	299.99	299.02	298.68	299.50	298.90	298.70
T_8 (K)	301.28	299.55	299.14	300.41	299.09	299.40
$\bar{h}$ W/(m ² K)	23.02	41.90	41.82 (Inv.)	29.22	54.73	52.61 (Inv.)

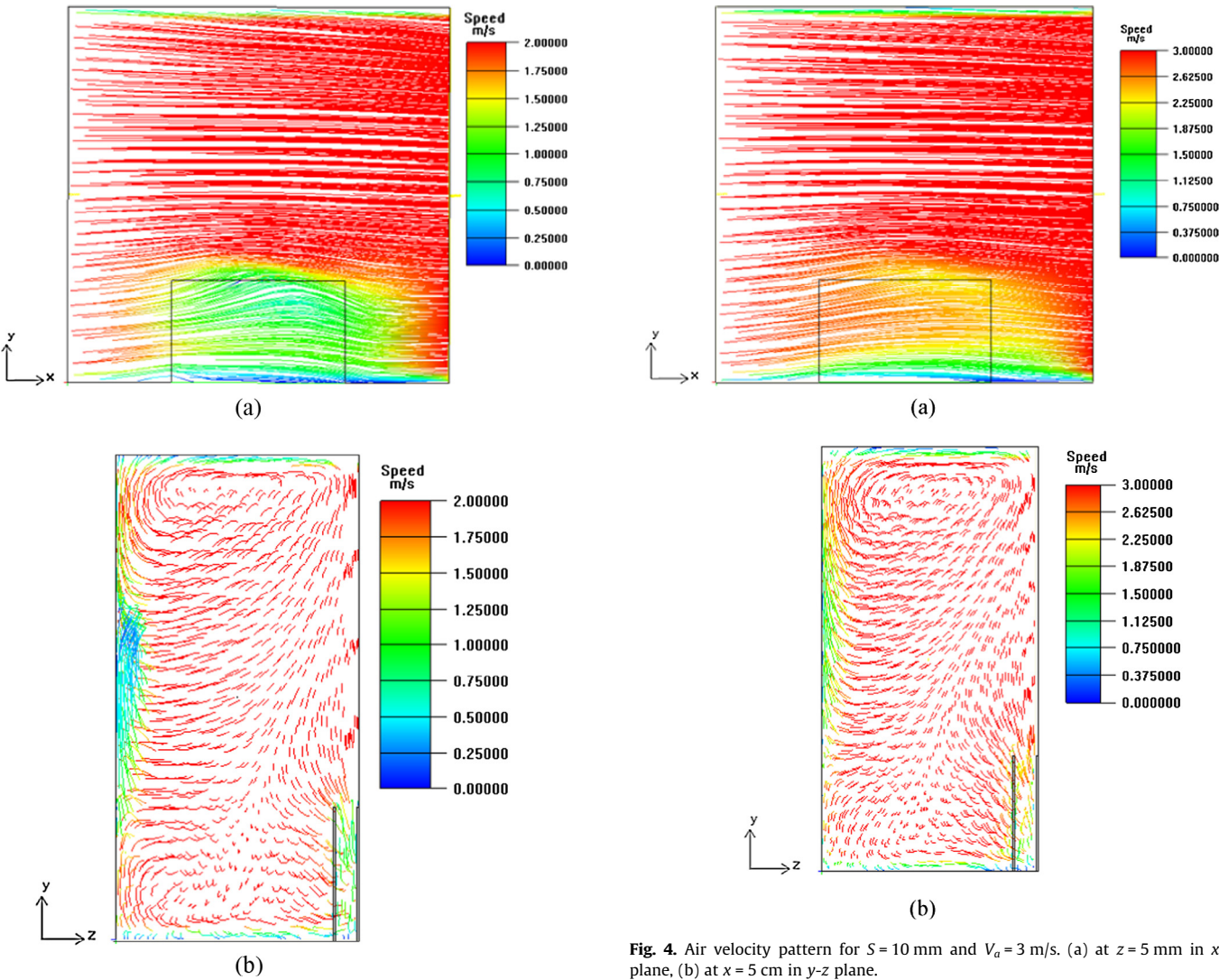


Fig. 3. Air velocity pattern for $S = 10$ mm and $V_a = 2$ m/s. (a) at $z = 5$ mm in x - y plane, (b) at $x = 5$ cm in y - z plane.

by correlation (32) and correlations (31) and (35) at $V_a = 1$ m/s, respectively. It can be seen in Ref. [24] that the fin height is 0.012 m. The fin height in this study is different from this value. In addition, the heat conduction equation of the fin is not solved. As can be seen in Ref. [21], the value of $\bar{h}$ decreases with decreasing fin height for a single fin. The present result of $\bar{h}$ is close to the

Fig. 4. Air velocity pattern for $S = 10$ mm and $V_a = 3$ m/s. (a) at $z = 5$ mm in x - y plane, (b) at $x = 5$ cm in y - z plane.

value of a single fin with $1 \text{ m/s} \leq V_a \leq 3 \text{ m/s}$ at $S = 20$ mm. The present results for $\bar{h}$, respectively, are 31.14 W/(m² K) and 35.13 W/(m² K) at $V_a = 1$ m/s with $H = 0.04$ m and 0.06 m. Therefore, the value of $\bar{h}$ can be estimated as 14.55 W/(m² K) with $H = 0.012$ m and $S = 5$ mm by extrapolation. It is seen that the present result of $\bar{h}$ is closer to that obtained by Eq. (35) for small fin spacing compared to the results shown in Table 2. However, the $\bar{h}_b$ results obtained by the inverse method and commercial software [16]

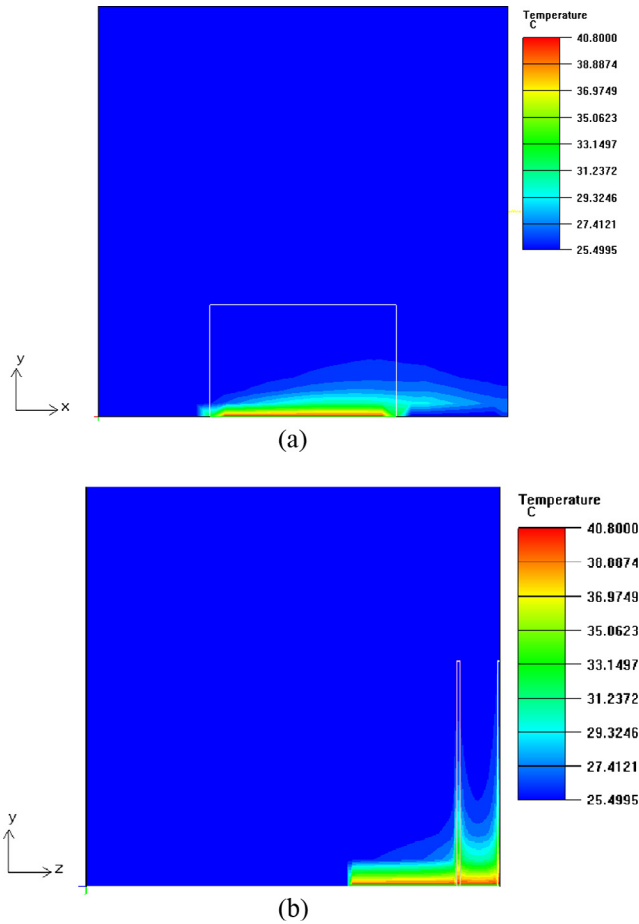


Fig. 5. Air temperature profile for $S = 10$ mm and $V_a = 2$ m/s. (a) at $z = 5$ mm in x - y plane, (b) at $x = 5$ cm in y - z plane.

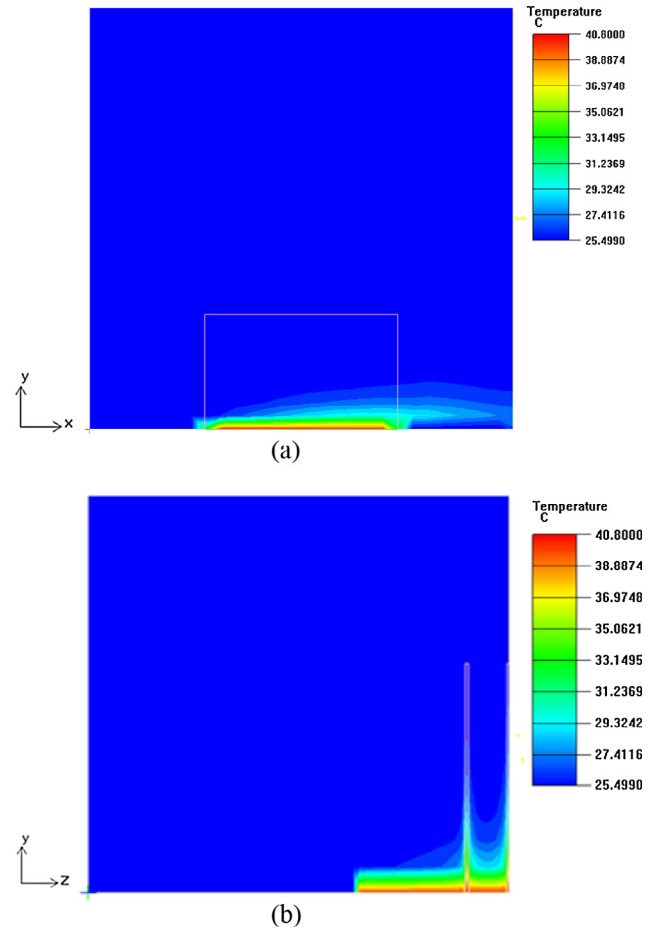


Fig. 6. Air temperature profile for $S = 10$ mm and $V_a = 3$ m/s. (a) at $z = 5$ mm in x - y plane, (b) at $x = 5$ cm in y - z plane.

are in good agreement with those obtained from the proposed correlation (34) with $1 \text{ m/s} \leq V_a \leq 3 \text{ m/s}$ and $5 \text{ mm} \leq S \leq 20 \text{ mm}$. Thus, the proposed correlation has good accuracy and reliability and can be more accurate than the correlations proposed by Elshahai [13] and Chen et al. [21].

Table 3 shows the effect of the total number of grid points N_t on the obtained results. The total number of grid points used to discretize the entire computational domain is taken from 61,479 to 693,924 for $V_a = 1 \text{ m/s}$, from 278,124 to 1,485,990 for $V_a = 2 \text{ m/s}$ and from 319,968 to 1,519,048 for $V_a = 3 \text{ m/s}$. It can be seen that in order to obtain more accurate results, the value of N_t should increase with increasing air velocity and fin spacing. The total number of grid points varies with fin spacing and air velocity. In other words, the number of grid points depends on fin spacing and air speed. This implies that the grid independence assumption shown in Refs. [4,9] may not be suitable for this problem.

In order to further validate the accuracy of the obtained $\bar{h}$ and $\bar{h}_b$ numerical results, a comparison of the fin temperature between the numerical results and the experimental data is required at each selected measurement location. Tables 4 and 5 show the comparison of the fin temperature at each selected measurement locations and the average heat transfer coefficient between numerical results obtained by the laminar and RNG k - ϵ turbulence models and the experimental data for various V_a values and $S = 5$ mm and 10 mm, respectively. It is seen in Tables 4 and 5 that the fin surface temperature obtained by the RNG k - ϵ turbulence model is closer to the experimental data than that obtained by the laminar flow model at each selected measurement location. The value

of $\bar{h}$ obtained by the RNG k - ϵ turbulence model is closer to the inverse result than that obtained by the laminar flow model. The important findings from Tables 1, 3 and 4 and Table 1 of Ref. [21] are that the values of $\bar{h}$, $\bar{h}_b$ and Q values obtained by FLUENT 4 are closer to inverse results than those obtained by FLUENT 15 along with SWF and EWT. However, the fin surface temperature obtained by both FLUENT 4 and FLUENT 15 is close to the experimental data. Therefore, a comparison of the heat transfer coefficient is also required in order to select the appropriate FLUENT version. The ratio between them is about 68.5% at $V_a = 2 \text{ m/s}$ with $S = 5 \text{ mm}$ and 54.9% with $S = 10 \text{ mm}$. This means that the RNG k - ϵ turbulence model is more suitable for this problem than the laminar flow model. However, laminar mixed convective heat transfer is assumed in Refs. [6,24]. The above results show that the flow model can vary with the physical geometry. Therefore, the choice of an appropriate flow model is important to determine more accurate numerical results.

Tables 4 and 5 show that the fins have a higher temperature drop near the fin base or from the fin base ($y = 0$) to $y = 3H/8$. T_j^{mea} and T_j^{num} at $x = L/4$ are lower than those at $x = 3L/4$ with various air velocities and fin spacings, especially near the fin base. The fin temperature in the downstream sub-region is higher than the upstream sub-region. In other words, the low velocity region occurs in the downstream sub-fin region.

The numerical results shown in Figs. 3–11 are obtained by the RNG k - ϵ turbulence model. Figs. 3 and 4 show the air velocity patterns in the x - y and y - z planes for $S = 10$ mm and $V_a = 2 \text{ m/s}$ and 3 m/s, respectively. The air in the wind tunnel is heated by the

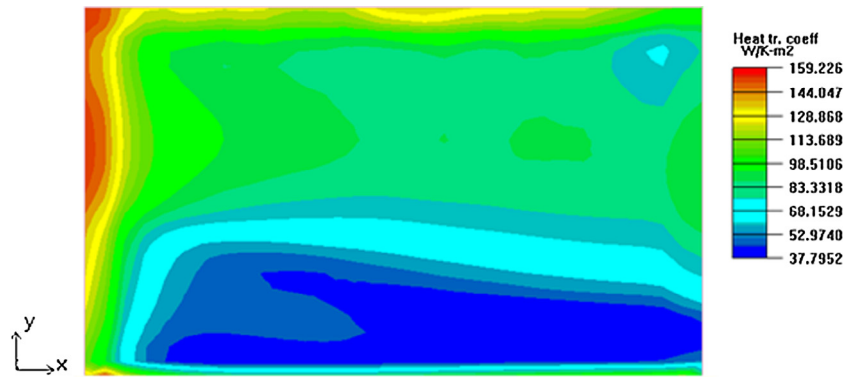


Fig. 7. Heat transfer coefficient distribution on the fins for $S = 10$ mm and $V_a = 3$ m/s.

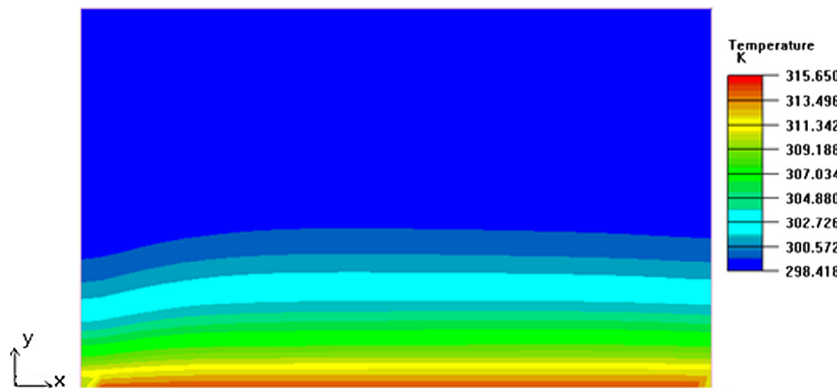


Fig. 8. Fin surface temperature profile for $S = 10$ mm and $V_a = 3$ m/s.

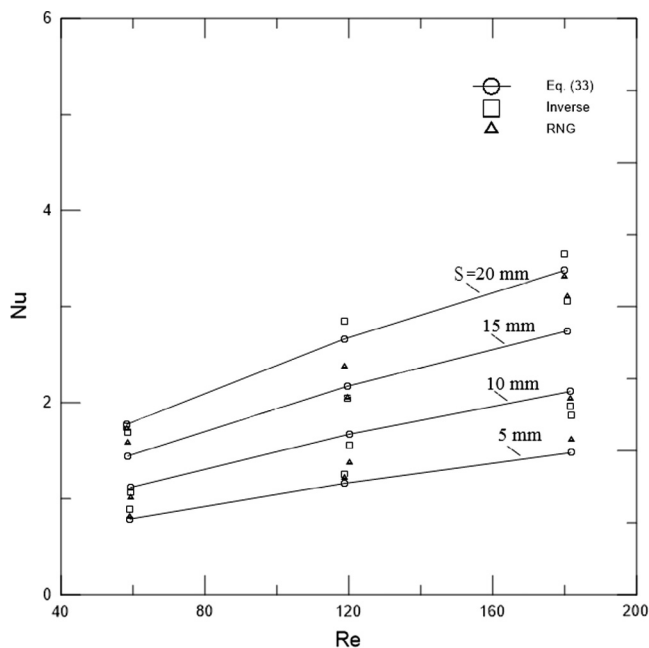


Fig. 9. Variation of Nu with Re for various values of S .

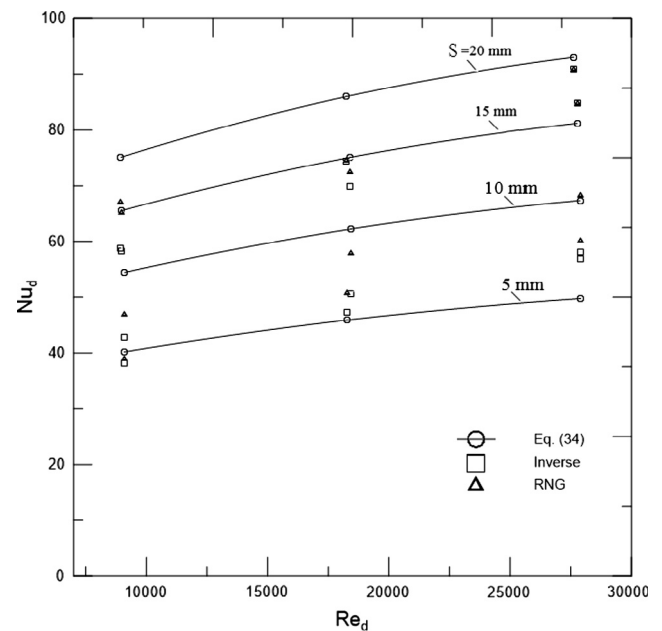


Fig. 10. Variation of Nu_d with Re_d for various values of S .

heated horizontal plate. The upward-moving heated air in natural convection is caused by two adjacent fins. Thus, the air flow may form a low velocity region near the tip of the fin. The Richardson number Ri and Rayleigh number Ra with $S = 10$ mm is $Ra = 3.62 \times 10^5$ (or $Ri = 6.68 \times 10^9$) for $V_a = 2$ m/s and

$Ra = 3.06 \times 10^5$ (or $Ri = 8.53 \times 10^9$) for $V_a = 3$ m/s. The flow field of this problem can be divided into three regions as shown in Figs. 3 (a) and 4(a). It can be seen from Figs. 3(a) and 4(a) that the air flow is closer to the uniform distribution with $V_a = 3$ m/s than with $V_a = 2$ m/s in the upstream region of the first section, except near the horizontal plate. As shown in Figs. 3(b) and 4(b), the secondary

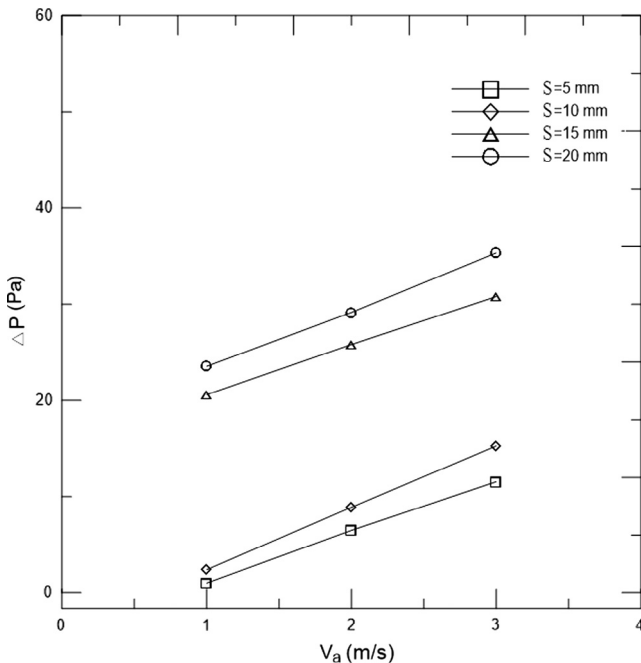


Fig. 11. Variation of Δp with V_a for various values of S .

vortices are found at two corners of the wind tunnel for $V_a = 2$ m/s. The low velocity region is observed at the sidewalls of the wind tunnel for $V_a = 2$ m/s and 3 m/s. Increasing the air velocity can reduce the low speed range, and the secondary vortex can be gradually destroyed in the lower corner of the wind tunnel at $V_a = 2$ m/s. Therefore, the secondary vortex is only found in the upper corner of the wind tunnel for $V_a = 3$ m/s.

The air flow between the shroud and the fin tip is more uniform at $V_a = 3$ m/s than at $V_a = 2$ m/s. The flow structure of the air flow entering the second section between two adjacent parallel fins contributes to the development of the boundary layer and is affected by the presence of the fin base and the free stream above the fins. As shown in Figs. 3(a) and 4(a), flow separation and reattachment are found near the fin base. The natural convection effect is more pronounced at $V_a = 2$ m/s than at $V_a = 3$ m/s. This means that the amount of cold air blown into the fins is smaller at $V_a = 2$ m/s than at $V_a = 3$ m/s. Thus, the complex 3D flow characteristics within the parallel fins are accompanied by upward-moving natural convection flow, flow separation and reattachment. The low velocity air flow is found in the third region of the downstream section from the fin near the horizontal plate. However, the scope of low velocity airflow is greater at $V_a = 2$ m/s than at $V_a = 3$ m/s in the y direction.

The air temperature profiles in the x - y and y - z planes, respectively, are shown in Figs. 5 and 6 with $S = 10$ mm for $V_a = 2$ m/s and 3 m/s. It is found in Figs. 5 and 6 that the air temperature profile in the x - y plane exhibits a non-uniform distribution in two adjacent fins. The thermal boundary layer thickness increases along the flow direction and decreases with increasing air velocity. The low temperature region is seen near the inlet region of the fin and the fin tip. A higher temperature and temperature gradient are observed near the fin base. This indicates that a high heat transfer rate exists in this region. The same results are found in Tables 4 and 5. The fin temperature measurements show a higher temperature drop in the vicinity of the fin base.

Figs. 7 and 8 show the heat transfer coefficient distribution on fins and the fin surface temperature profile for $S = 10$ mm and $V_a = 3$ m/s, respectively. As shown in Figs. 3(a) and 4(a), the air velocity is lower in the downstream region of the fin than in the

upstream region of the fin. Therefore, the difference between the fin temperature and the ambient air temperature is lower in the upstream region of the fin than in the downstream region of the fin. It is found from Fig. 7 that the heat transfer coefficient is higher in the upstream region of the fin than in the downstream region of the fin. The results are similar to those shown in Table 1. It can be seen from Fig. 8 that the temperature of the fin in the vicinity of the fins base is higher than the other regions. The temperature distribution in the upper region of the fin is approximately uniform. Similar results can be found in Table 1 of Ref. [21] and Tables 4 and 5. Thus, the present results are reliable.

The variation of Nu with Re is shown in Fig. 9 for various values of S . Fig. 10 shows the variation of Nu_d with Re_d for various S values. It is found from Figs. 9 and 10 that Eqs. (33) and (34) are in good agreements with FLUENT 4 and inverse results. This means that Eqs. (33) and (34) are reliable. The Nu and Nu_d values increase with increasing S value for $5 \text{ mm} \leq S \leq 20 \text{ mm}$ at the same air velocity.

Fig. 11 shows the variation of the pressure drop Δp with V_a for various S values. It is seen that the Δp value increases with increasing values of S and V_a . The Δp value has a greater variation from $S = 10$ mm to 15 mm for $1 \text{ m/s} \leq V_a \leq 3 \text{ m/s}$. The present results agree with those shown in Fig. 11 of Ref. [27]. The change in pressure drop is small for $5 \text{ mm} \leq S \leq 10 \text{ mm}$ and $15 \text{ mm} \leq S \leq 20 \text{ mm}$. A similar phenomenon can also be found in Ref. [27].

7. Conclusions

This study uses commercial software along with inverse results, appropriate flow model, wall function and experimental data to determine the mixed convection fluid flow and heat transfer characteristics of plate-fin heat sinks. The results show that FLUENT 4 has better accuracy than FLUENT 15 along with SWF and EWT. The total number of grid points needs to be increased with increasing air velocity and fin spacing to obtain more accurate results. This means that the grid independence assumption may not be appropriate. Therefore, the impact of the flow model, near-wall treatment, FLUENT version and grid point on the results obtained cannot be ignored. The $\bar{h}$ and $\bar{h}_b$ results obtained by the RNG k - ϵ turbulence model are in good agreement with the inverse results. The resulting fin temperatures are close to the experimental data at each selected measurement location. Importantly, the RNG k - ϵ turbulence model is more suitable for this problem than the laminar flow model. The two proposed correlations are close to the inverse and FLUENT 4 results. The secondary vortices are found in the two corners of the wind tunnel for $V_a = 2$ m/s. However, the secondary vortex only appears in the upper corner of the wind tunnel for $V_a = 3$ m/s. The pressure drop increases with increasing values of S and V_a . In order to correctly understand the fluid flow and heat transfer characteristics of the plate-fin heat sinks obtained from commercial software, a comparison of $\bar{h}$, $\bar{h}_b$ and the fin temperature at each selected measurement location is required.

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